Digital Communication IC Design

Homework #1

(Due on 2026/10/7)

- Please upload your codes and reports (with simulation results and required explanations) to the NTUCool.
- Please disclose how you use AI tools and its version.

Total points: 130 points

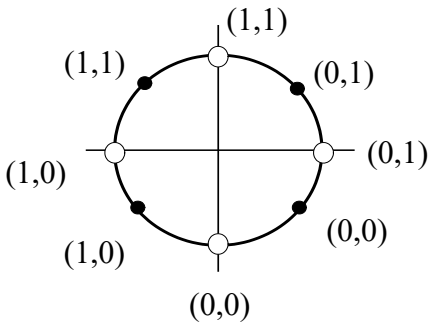
- Please randomly generate 32 binary bits by using your last two digits as the seed, and **print out** your own bits $(b_i, i = 0,1, \dots 31)$. These binary data will be used throughout homework 1. TA will use your binary bits to check your answer. So, don't change your data for all the questions.
- Use the following settings for single-carrier systems in Q1.

Quantity	Symbol	Value	Derived interpretation
Symbol period	T_{sym}	0.1 μs	Symbol rate = 10 M symbol/s
Carrier frequency	f_c	40 MHz	4 carrier cycles per symbol
Sampling frequency	f_s	640 MHz	16 samples per carrier cycle
Samples per symbol	-	64	Half-symbol delay = 32 samples

1. Define

$$p(t) = \begin{cases} 1, & 0 \le t < T_{sym} \\ 0, & \text{otherwise} \end{cases}$$

and generate the complex baseband QPSK signal by using the mapping of the black point defined in Fig. 1 as a_k, where the constellation points lie on the unit circle.



(b_{2k}, b_{2k+1})

Fig. 1

(a) Plot its real and imaginary parts. Note that the x-axis should be expressed in μs and t

start from 0. (5%)

$$x_{BB,QPSK}(t) = \sum_{k=0}^{15} a_k p(t - kT_{sym})$$

(b) Upconvert the baseband waveform to the real passband signal as defined in the course slides. Plot the first eight symbol periods and mark all symbol boundaries with vertical black dashed lines. (10%)

(c) On a separate enlarged figure, show the baseband and passband signals over the first two symbol periods. If both appear in one figure, identify their amplitude scales and the time scales clearly. (5%)

(d) Use an FFT to plot the magnitude spectra of the baseband and passband signals. Use a frequency axis in MHz. Note that the range of y-axis should be less than 60dB. (10%)

(e) Generate O-QPSK baseband signal, $x_{BB,OQPSK}(t)$, by delaying the Q branch by $T_{sym}/2$. Assume $Q(t) = I(t)$ for $0 \le t < T_{sym}/2$ (5%).

(f) Draw the figure of phase jump in the range of $[-\pi, \pi)$ versus $0 \le t < 16T_{sym}$ for your O-QPSK baseband signal by using "stem". (10%)

(g) What's your observation compared to the phase jump of QPSK baseband signal in Q1(a). (5%)

2. Use b_0, ..., b_{31} and QPSK mapping of white points for X_k in Fig. 1 on all $N = 16$ subcarriers. This is a baseband-only exercise. Define the baseband sample interval $T_s = 25\ ns$

(a) Calculate the useful OFDM symbol duration T_{sym}, subcarrier spacing Δf.(4%)

(b) Create a table containing $k, (b_{2k}, b_{2k+1}), X_k$, and the signed subcarrier frequency. Use $f_k = k\Delta f$ for $k = 0, \ldots, 7$ and $f_k = (k - N)\Delta f$ for $k = 8, \ldots, 15$. (10%)

(c) Let $t = uT_s/(16)$ for $u = 0,1,2,\ldots 16N$. Generate the individual subcarrier waveforms $y_k(t)$ versus t for $k = 1$ and 14. Plot their real and imaginary parts and verify that each subcarrier completes the expected number of cycles during T_{sym}. **Please state how you verify it** (16%)

$$y_k(t) = X_k e^{j2\pi f_k t}, 0 \le t < T_{sym}$$

(d) Sum all 16 subcarrier waveforms. Plot the real part, imaginary part of $y(t)$ versus t.

(5%)

$$y(t) = \sum_{k=0}^{N-1} X_k e^{j2\pi f_k t}$$

(e) Now, calculate x[n] using the 16-point IFFT definition below. Please use "stem" to draw the discrete samples $Nx[n]$ and plot $y(t)$ in the same figure for real-part and imaginary part. (10%)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j2\pi \mathrm{kn}/N}, n = 0, \dots, N-1$$

(f) In Q2(b) and Q2(d), we use $f_k = (k-N)\Delta f$ for $k \geq 8$, but for generating $x[n]$, $k\Delta f$ is assumed for $k \geq 8$. Although we use different frequency values, actually the discrete samples $Nx[n]$ have the same magnitude of the real part and imaginary part as those of $y(nT_s)$. Why? (5%)

3. Now, let's generate the OTFS baseband signals. Assume that $N = 4, M = 8$. Use the mapping of $X_{DD}[l,k] = 1 - 2b_i$, where $0 \leq l \leq M-1,\ 0 \leq k \leq N-1$, and $i = kM + l$.

(a) First, apply ISFFT to change the signal from delay-Doppler domain to time-frequency domain. Plot the real-part and imaginary part of the time-frequency grid $X_{TF}[0,m]$ and $X_{TF}[2,m]$ with labeled subcarrier index m . (10%)

$$X_{TF}[n,m] = \frac{1}{\sqrt{NM}} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} X_{DD}[k,l] e^{j2\pi(\frac{nk}{N} - \frac{ml}{M})}$$

(b) Next, change the signal from time-frequency domain $X_{TF}[n,m]$ to delay-time domain $X_{DT}[n,l]$ by applying unitary IFFT along the frequency domain subcarrier index m, just like your OFDM system, and repeats for every time index n with $n = 0,1,2,3$. Then re-arrange the delay-time domain signal into one-dimensional time-domain signal $x[q]$, where $q = nM + l$. Plot the real part and imaginary part of $x[q]$. (10%).

(c) How do you know whether your result is correct or not? Please show your verification. (10%)